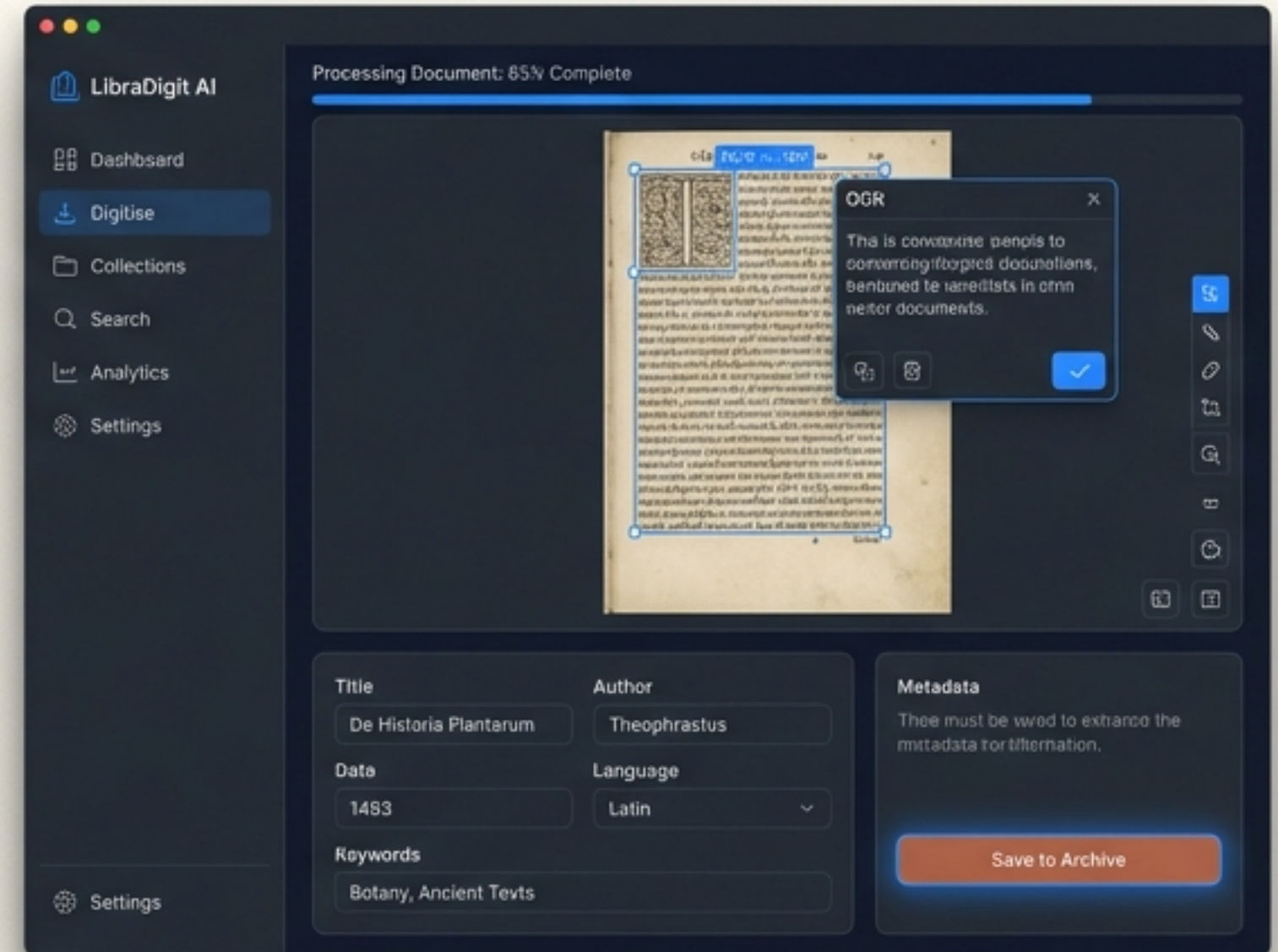


LibraDigit AI: The Future of Library Digitisation

A production-grade, offline-first suite for converting physical documents into searchable, metadata-rich digital archives.



Version 1.1.0

License: MIT (Open Source)

Platform: Desktop (Electron)

Built with ❤️ for librarians and archivists worldwide.

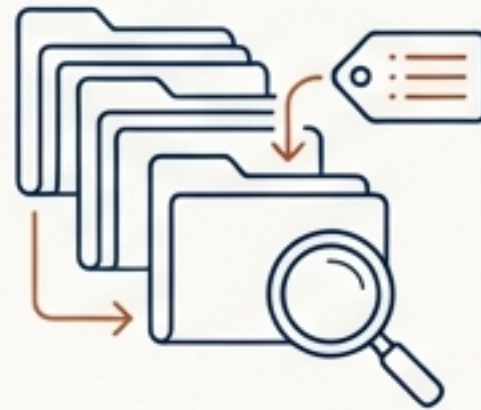
A Complete Solution for the Digital Archive Lifecycle

LibraDigit AI is a desktop application designed to streamline the preservation of institutional knowledge without relying on cloud infrastructure.



Digitise

Convert images and PDFs using Advanced OCR with AI-powered layout analysis and handwriting recognition.



Organise

Enrich files with comprehensive metadata (Author, Subject, Year) and XMP embedding.



Secure

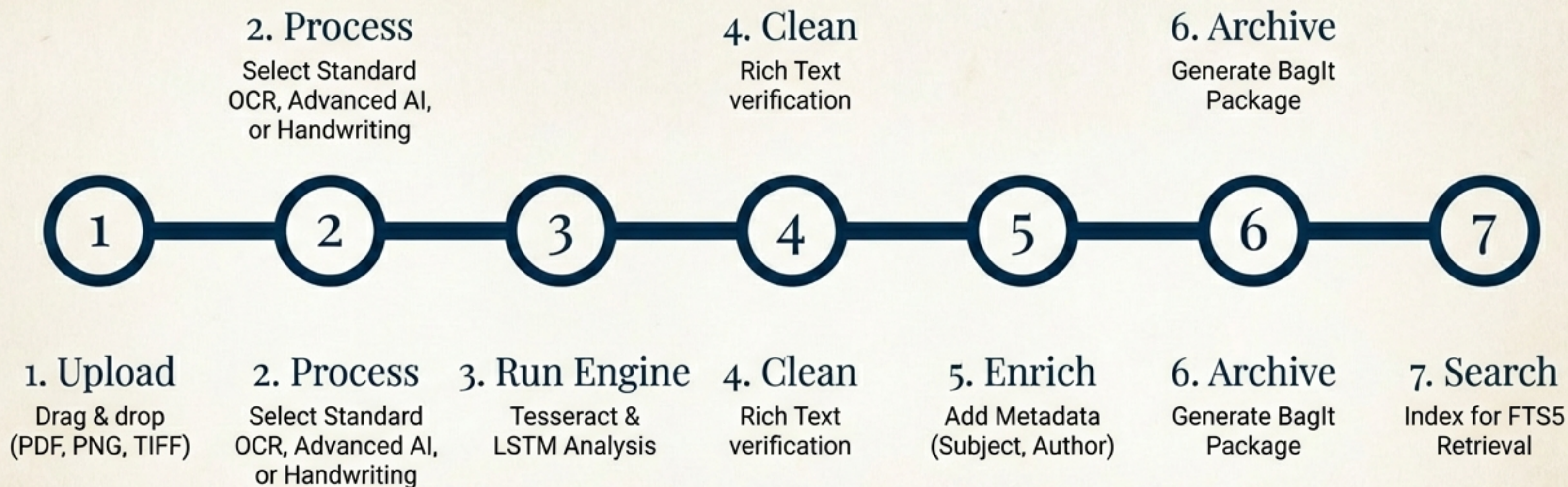
Operate with zero cloud dependency. All data, hashes, and files remain on the local machine.



Analyze

Monitor digitisation progress via a built-in statistics dashboard and full-text search engine.

The Digitisation Workflow



Advanced OCR & Intelligent Layout Analysis

Beyond simple text extraction, LibraDigit AI understands the structure of your documents.

-  **Layout Understanding:** Detects headers, footers, official stamps, and signatures.
-  **Structured Data:** Identifies tables and extracts content; detects form fields and checkbox status.
-  **Image Preprocessing:** CLAHE enhancement, adaptive thresholding, and denoising.
-  **Auto-Correction:** Fixes page rotation (0°, 90°, 180°, 270°) automatically.

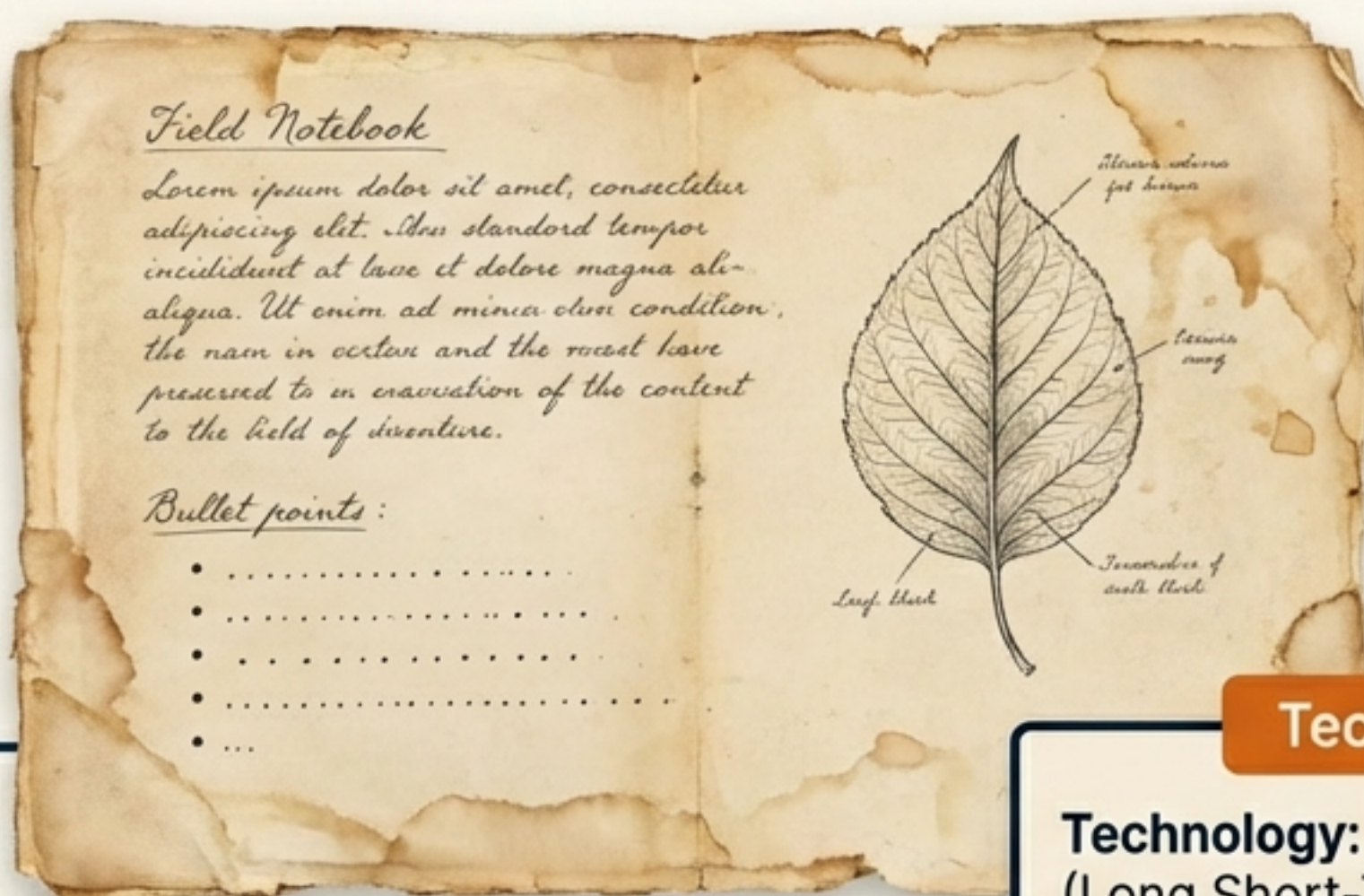


The illustration shows a document with several analysis overlays:

- Header Detected:** A blue label at the top left.
- OFFICIAL:** Two labels flanking a central seal at the top.
- Form Fields:** Labels for "Seatore of Name", "Bmael Naroo", "Age of Address", and "Date of".
- Structured Data Extraction:** A green label above a table.
- Table:** A table with 6 columns: City, Name, Taos, Rio, Mattie, and Eriots. It has 5 empty rows.
- Checkbox Status:** An orange label with a checked checkbox icon.
- Auto-Correction:** An orange label with a circular arrow icon and the text "Auto-Correction: 2° CW".
- Signature Block:** A purple label above a signature.
- Seal:** A circular seal at the bottom left.

Transforming Handwritten Notes into Searchable Data

Raw Input



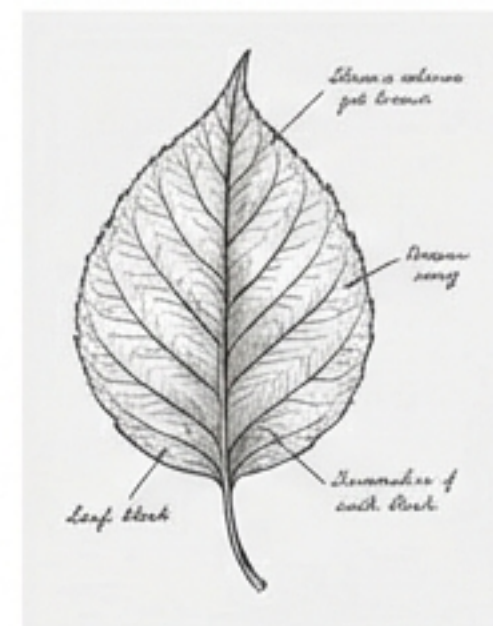
Processed PDF

Field Notebook

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua, enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi: ut aliquip ex ea commodo consequat.

Bullet points

- Heading, paragraph, lists.
- Terminas and technical content.
- Ideal for archiving manuscripts.



Technology

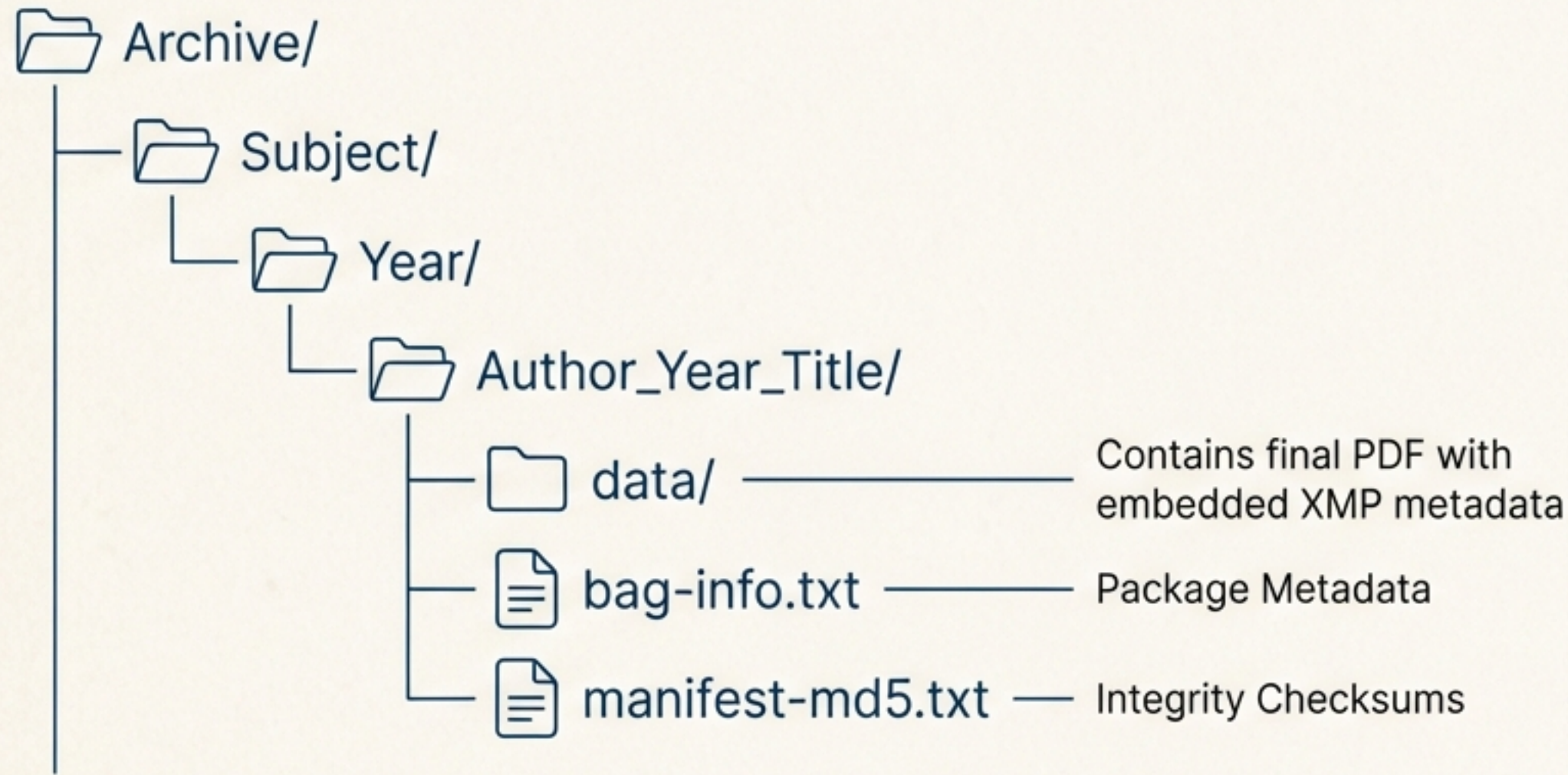
Technology: Specialized LSTM (Long Short-Term Memory) neural networks.

Accuracy: 75–92% on cursive and hand-printed text.

- **Retains structure:** headings, paragraphs, and lists.
- **Preserves** diagrams and technical content.
- **Ideal** for archiving field logs and manuscripts.

Preservation Standards and Data Integrity

Documents are organised using the international BagIt standard, ensuring verifiable data packages that remain uncorrupted over decades.



Key Features



XMP Embedding

Metadata (Title, Author, Keywords) travels inside the PDF binary.



BagIt Packaging

International standard for verifiable transfer.



Integrity

Automatic MD5 checksum generation.

Instant Discovery with Full-Text Search

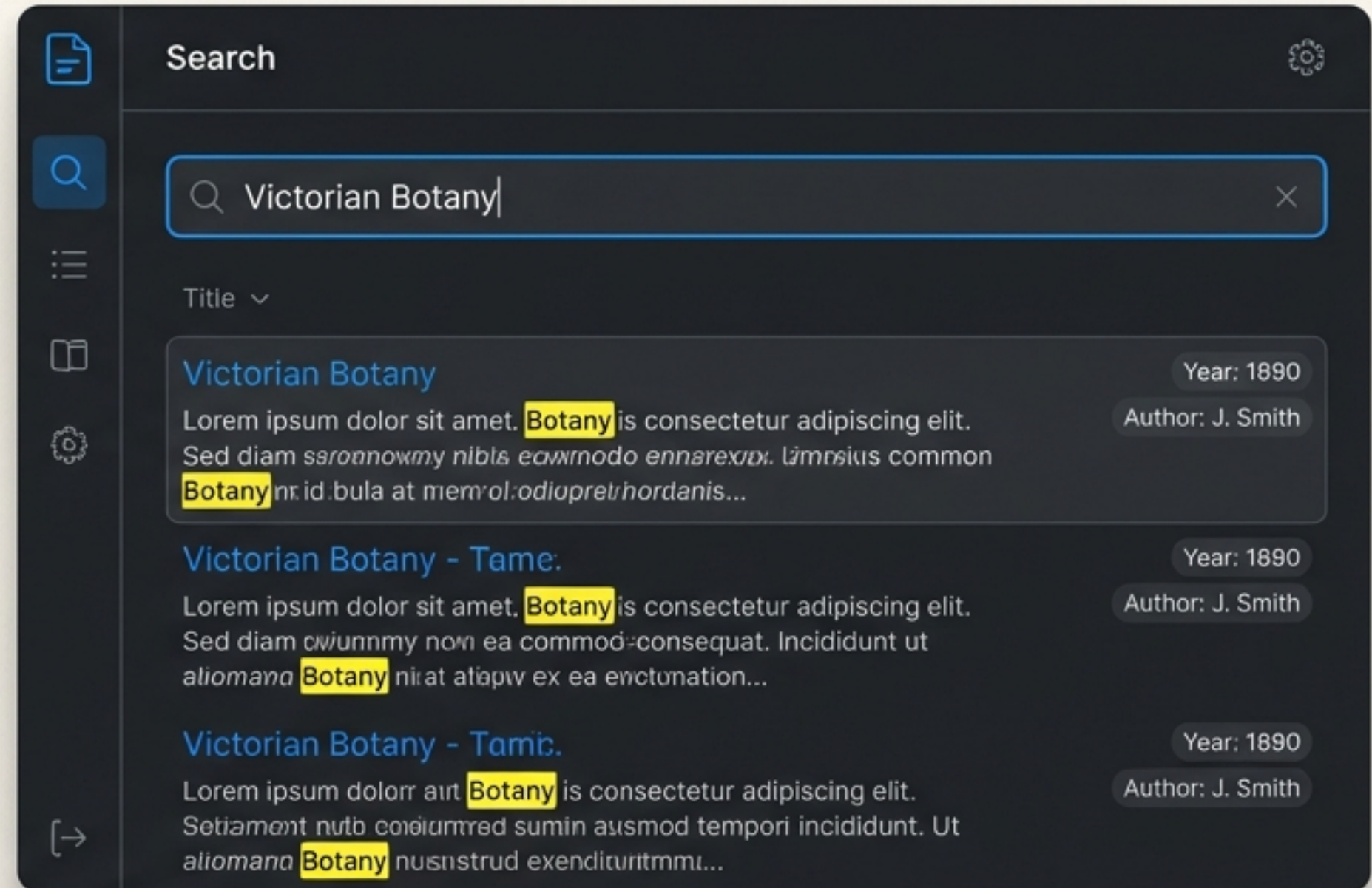
Search across the entire archive instantly using a dedicated engine.

Powered by **SQLite's FTS5** extension.

Deep Retrieval: Search thousands of documents.

Universal Scope: Query by Title, Author, Keywords, or full body text.

Context-Aware: Results display exact context with highlighting.



Project Analytics and Dashboard Insights



Track Progress:

Monitor files across all workflow stages.



Storage Management:

Real-time disk usage monitoring.



Team Output:

Visualize productivity trends.



Content Analysis:

Track top subjects in the archive.

Workflow Distribution

Upload

OCR

Cleanup

Metadata

Archived



Weekly Productivity



Storage Metrics

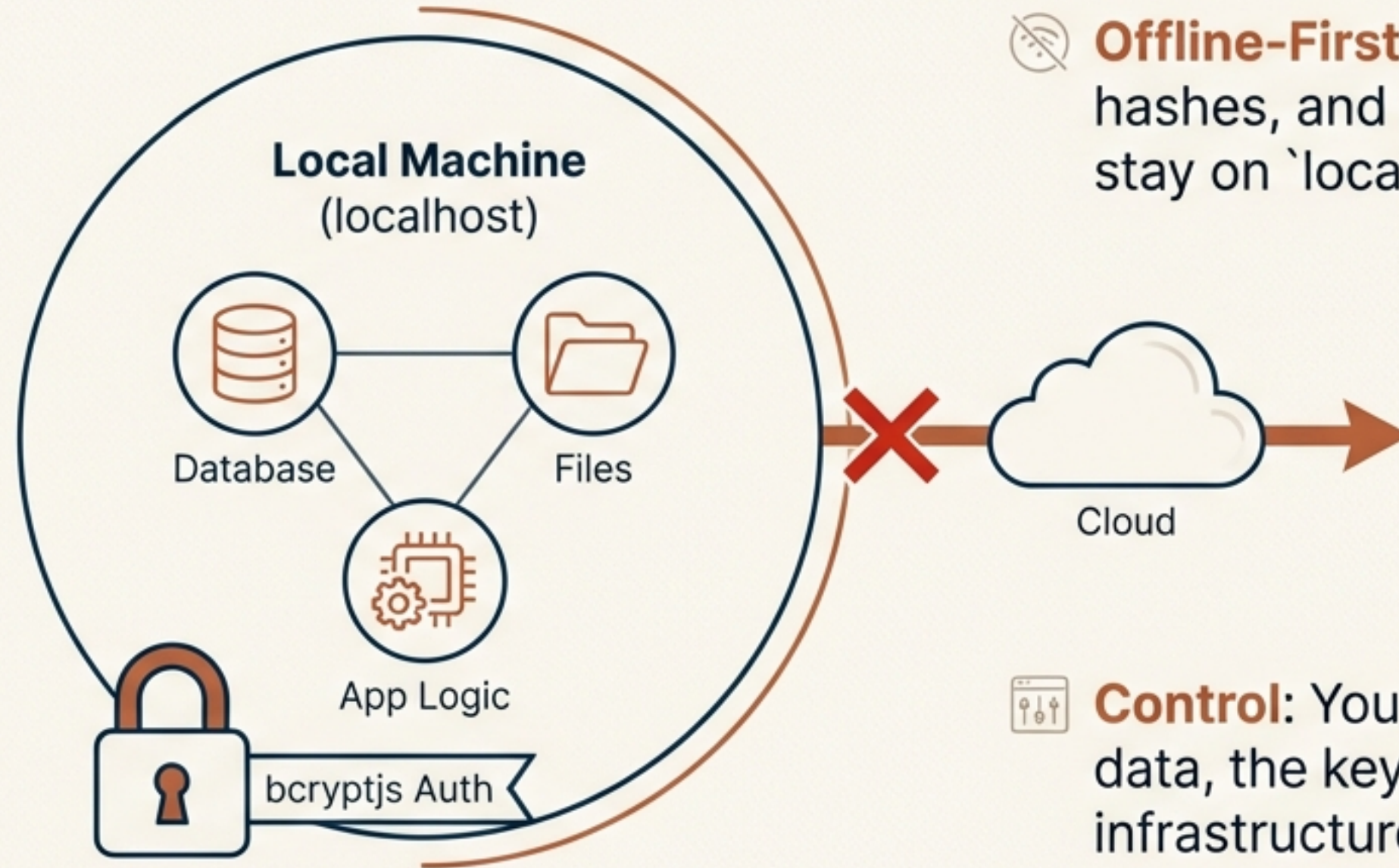


Security and Privacy Architecture

 **Privacy-First:** Zero cloud dependency.

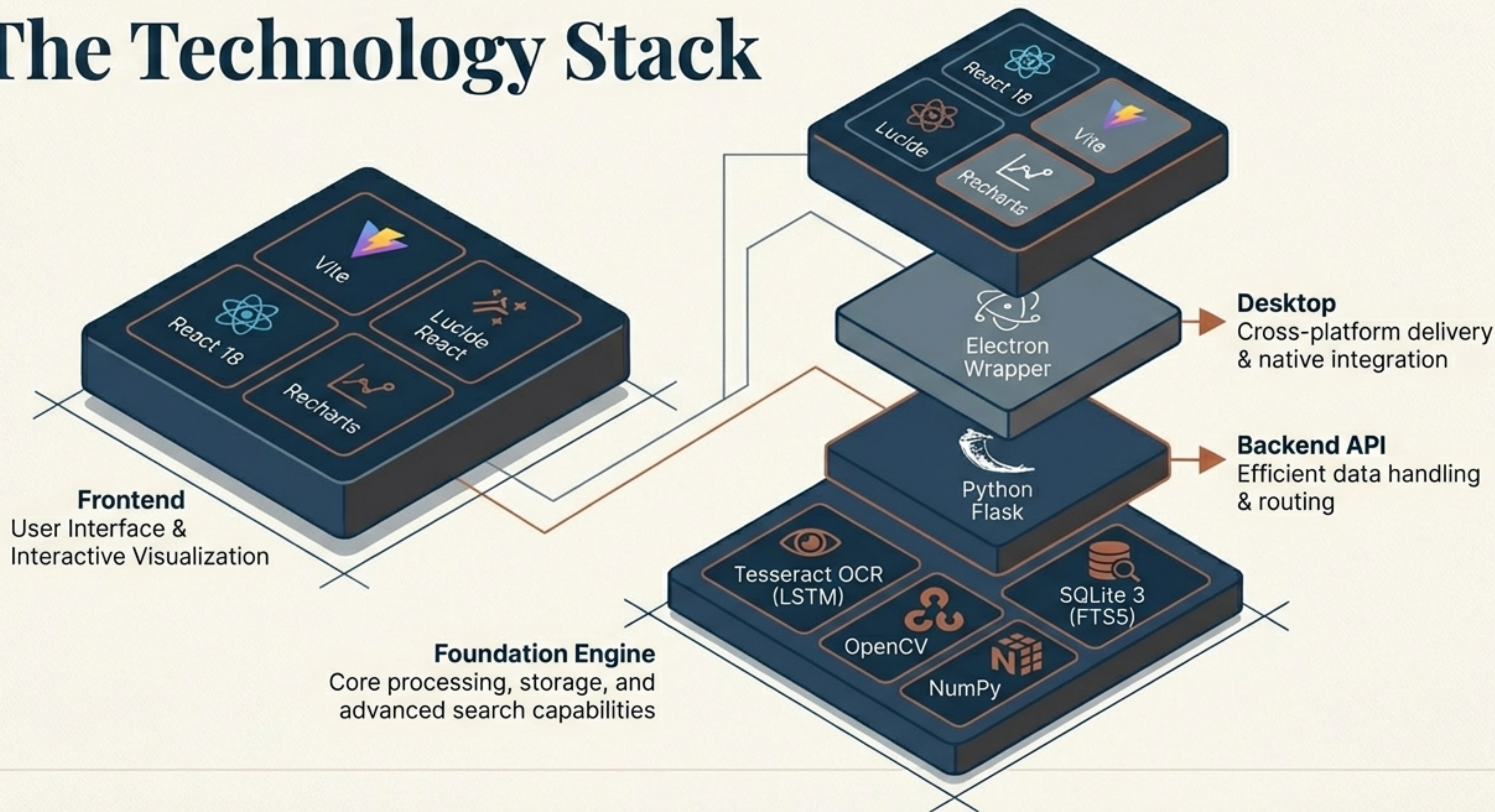
 **Offline-First:** All data, hashes, and archives stay on `localhost`.

 **Secure Auth:** Local password management with hashing.



 **Control:** You own the data, the keys, and the infrastructure.

The Technology Stack



Installation and Prerequisites

System Requirements

- **Node.js:** v18 or higher
- **Python:** v3.8 or higher
- **Tesseract OCR:** Required for extraction (Win/Mac/Linux)

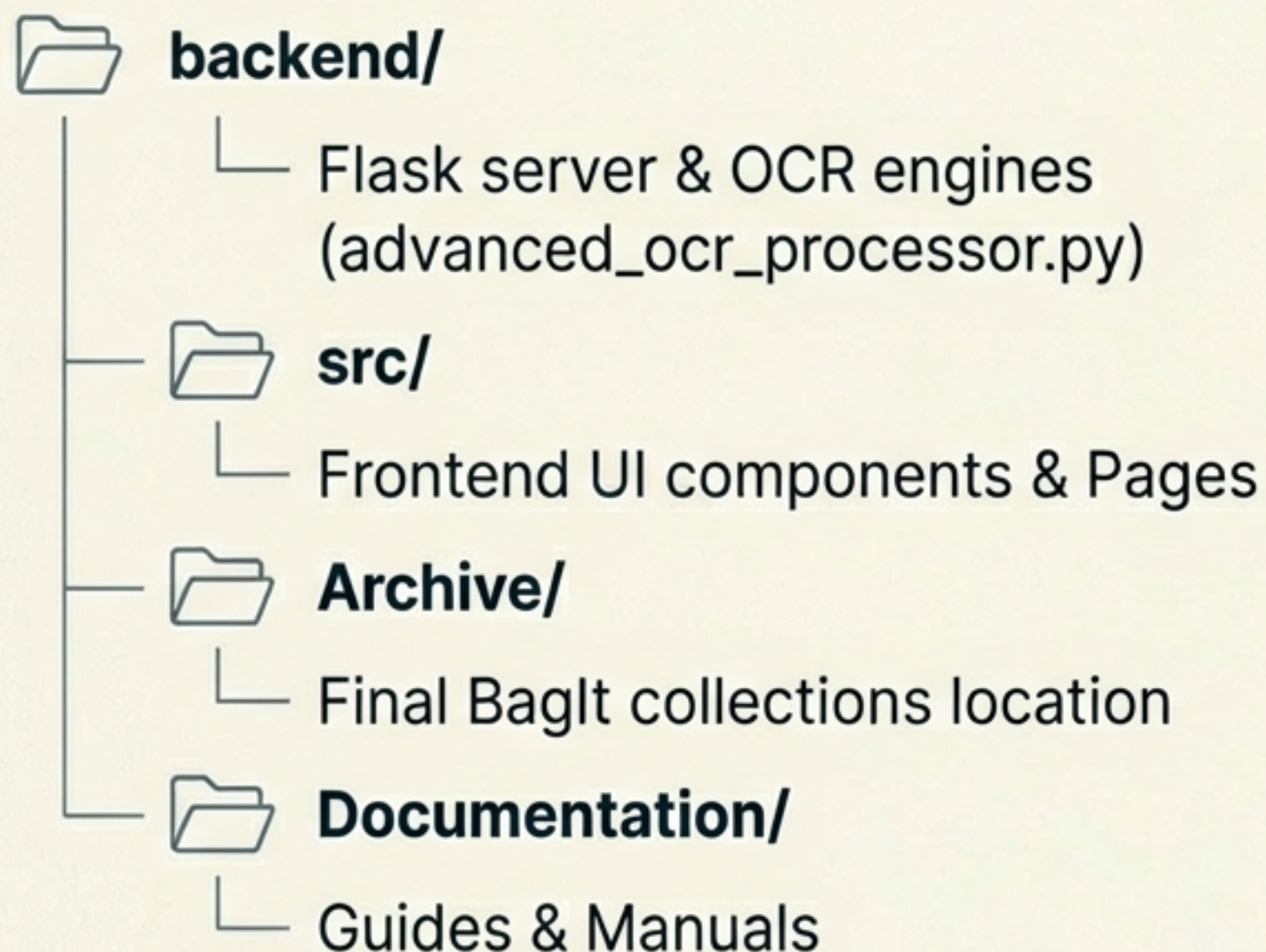
```
# Clone Repository
git clone [repo_url]

# Install Frontend
npm install

# Install Backend
pip install -r requirements.txt

# Launch
npm run dev
```

Project Structure and Documentation



- **Modular Design** separating logic for OCR and Metadata.
 - **Visual Install Guide:**
public/install_guide.html
 - **Advanced OCR Docs:**
ADVANCED_OCR_DOCUMENTATION.md
 - **Handwriting Guide:**
HANDWRITTEN_TO_PDF_DOCUMENTATION.md

Start Building Your Digital Archive

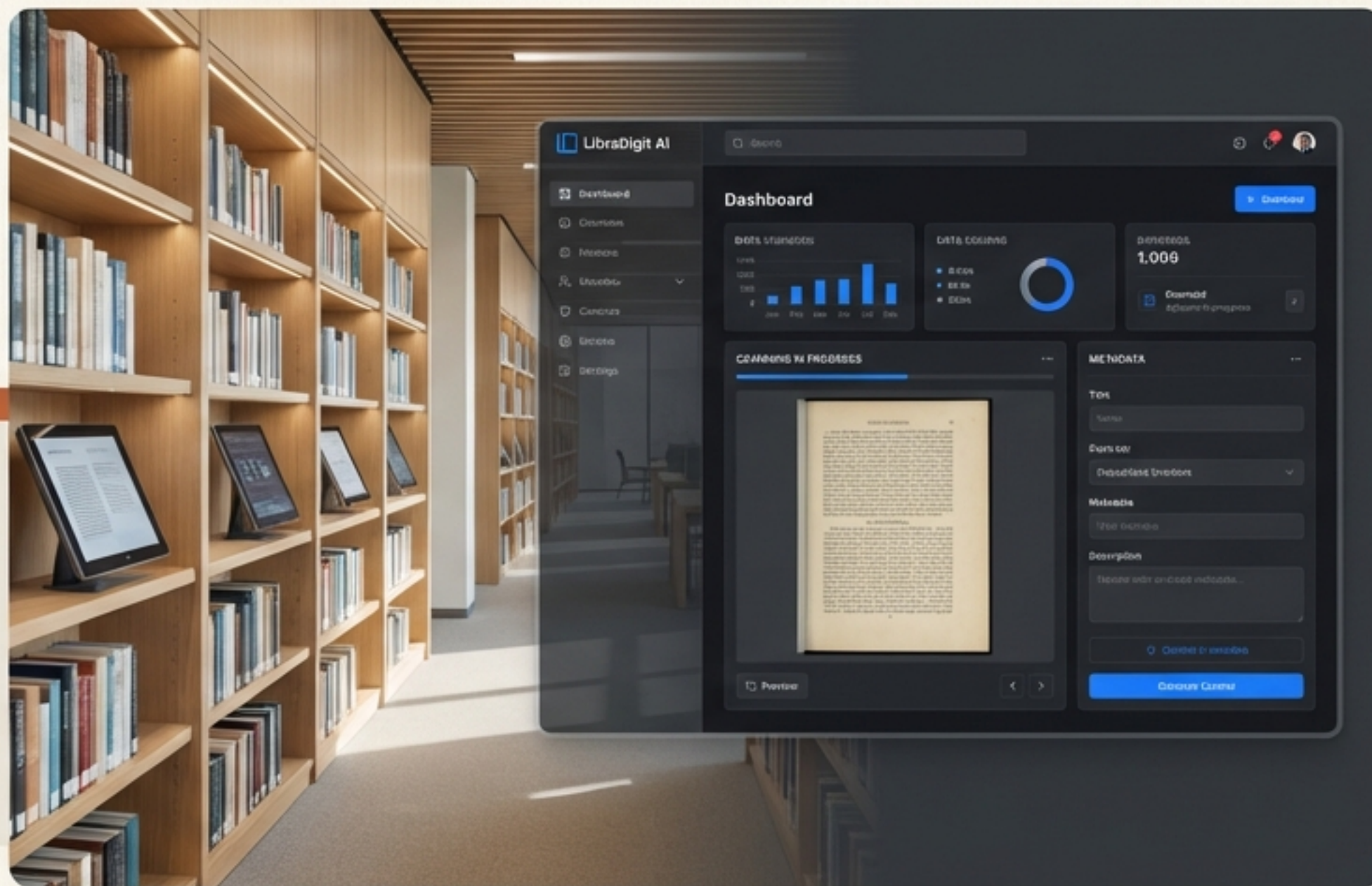
LibraDigit AI combines the power of computer vision with the rigour of archival standards to preserve history.

Download Source:

github.com/carthworks/LibraDigit-AI

License: MIT Open Source

Action: Follow the Quick Start guide to launch your first project today.



Get Started



Scan to view on GitHub

github.com/carthworks/LibraDigit-AI